START HERE

Your guide to cost and quality

Schwalbe, ninth edition: Chapter 7 pp. 285-323; Chapter 8 pp. 327-370. Both supplied lecture decks.

ITEC 3505 - Chapters 7 and 8

This guide explains the new lecture material in plain language, using the assigned textbook as the main authority. Chapter 7 asks whether the project is using money effectively. Chapter 8 asks whether its processes and results satisfy stakeholder needs. The two connect: preventing defects may increase early spending while reducing the total cost of delivery and support.

What you should be able to do

- Distinguish cost plans, estimates, baselines, funding needs, and forecasts.
- Calculate earned-value measures and explain what their signs and ratios mean.
- Choose a quality process, tool, or test level for a situation.
- Explain prevention, failure costs, Six Sigma, and process improvement.

Use this with your website

The **Chapters 7-8** tab contains **60 new questions**: 24 MCQs and 6 true/false questions for each chapter. It includes practice, flashcards, chapter filters, timed practice, and explained review. The separate printable question page includes all 60 questions and their answer key. The mix, timing, and equal chapter allocation are study choices; your instructor's assessment format and weighting for these chapters have not been confirmed.

How to study

Read an explanation, cover it, then explain the idea in your own words. Work the numerical examples without looking at the solution. In the website, filter to your weaker chapter and retry missed questions. Read the cited textbook section whenever you cannot explain why a distractor is wrong.

Source convention: page references mean printed textbook pages. Add 25 for the page shown by the supplied textbook PDF viewer. Slide references are PDF slide numbers in the new Chapter 7 or Chapter 8 lecture file.

Example convention: examples and calculations are original teaching examples unless labelled **Textbook case**. Summaries paraphrase the sources. Historical statistics and software editions in the book describe its publication period, not current conditions.

CHAPTER 7

Cost management: four different jobs

Textbook pp. 286-289, 294, 305-307; slides 4-8, 15, 26-28.

Cost is the value of resources given up to achieve an objective. Money is the usual measurement, but internal staff time and resources still have a cost even when no new supplier invoice arrives. Project cost management aims to complete the defined project within an approved budget; minimizing every expense regardless of consequences is not the objective.

Process and group	Main question	Important outputs
Plan Cost Management - Planning	How will we estimate, report, and control costs?	Cost management plan
Estimate Costs - Planning	What resources and money will the work require?	Cost estimates; basis of estimates; document updates
Determine Budget - Planning	When and where will the approved money be allocated?	Cost baseline; funding requirements; document updates
Control Costs - Monitoring and Controlling	How are we performing, and what changes need action?	Performance information; forecasts; change requests; plan/document updates

Read Figure 7-1 as a flow of decisions

The charter and project plans establish context. Estimates draw on scope, resource needs, historical data, judgment, and uncertainty. Budgeting aggregates and time-phases estimates. Control compares actual progress and spending with the approved baseline. Outputs feed later decisions and may require authorized changes; the processes can be revisited.

What belongs in a cost management plan?

Specify accuracy and rounding, units such as person-hours, accounting/control-account codes linked to the WBS, thresholds for action, earned-value measurement rules, report formats and frequency, and the procedures for each cost process. A control account is a management/accounting point for accumulating and tracking costs.

Example: a rule requiring investigation when CPI falls below 0.90 belongs in the plan. A $100,000 approved month-three cumulative budget belongs in the baseline. A new $125,000 forecast belongs in cost reporting and forecasting. These documents serve different purposes.

Why this matters: IT cost overruns can reflect unclear requirements, premature estimates, integration complexity, and weak change control. The book's NHS and high-overrun examples illustrate risk and governance problems; their dates and dollar statistics are historical case information.

CHAPTER 7

Financial language and cost classifications

Textbook pp. 290-293; slides 9-14.

Profit = revenue - expenditures. Profit margin = profit / revenue. A company with $500,000 revenue and $450,000 expenditures earns $50,000 profit and a 10% margin. If the same margin applies to additional sales, another $20,000 of revenue produces $2,000 of profit, not $20,000.

Life cycle costing considers development plus operations, maintenance, and support over the product's useful life. A low purchase price can conceal expensive licensing, training, downtime, or support. **Cash flow analysis** examines when costs and benefits occur; an attractive total return does not solve a shortage of cash in the month invoices fall due. State whether amounts use constant-year or future-year dollars.

Distinction	Meaning	Original example
Tangible / intangible	Easy / difficult to express reliably in money	$8,000 license invoice / improved organizational reputation
Direct / indirect	Attributable to this project / allocated shared overhead	Dedicated developer hours / shared office electricity
Sunk / future	Already incurred / still affected by a decision	Paid feasibility study / optional next development phase

These classifications are independent. A dedicated license can be both tangible and direct. Intangible does not mean unimportant, and indirect does not mean free. **Cost overrun** is the excess over the estimate; percentage overrun = (actual cost - estimated cost) / estimated cost x 100%. A $125,000 result against a $100,000 estimate is a $25,000 or 25% overrun.

Sunk-cost decision rule

A prototype has already cost $90,000. Finishing would require another $40,000 and generate only $25,000 of future benefits. Under these assumptions, continuing loses another $15,000. The $90,000 already spent is not recovered by approving more work. Consider genuine future consequences, such as contractual termination costs, separately.

Learning curves and reserves

For repetitive production, **learning-curve theory** predicts a regular decline in unit effort or cost as experience accumulates. Do not assume every novel task benefits equally. **Reserves** address uncertainty: contingency covers identified risks or known unknowns; management reserve covers unforeseen work or unknown unknowns within the project context. Their different budget treatment is explained next.

Study check: why can early requirements work be financially worthwhile? It can avoid much larger downstream correction and support costs, improving the total life cycle outcome.

CHAPTER 7

Estimate types and estimation methods

Textbook pp. 294-299; slides 16-21. Accuracy ranges follow this edition.

Estimates become better informed as scope and technical understanding improve. Their purpose, uncertainty, and date matter; a single early number should not silently become a permanent promise.

Type	Usual purpose and timing in the text	Typical accuracy
Rough order of magnitude (ROM)	Early selection/screening; often 3-5 years before completion	-50% to +100%
Budgetary	Allocate funds in advance; often 1-2 years out	-10% to +25%
Definitive	Detailed purchasing and final-cost decisions; generally within a year	-5% to +10%

A $100,000 ROM therefore spans $50,000-$200,000 under these typical bounds. These are textbook ranges, not guarantees or universal rules for every industry. Other industries use additional estimate classes.

Method	How it works	Main strength / caution
Analogous, or top-down	Adapt the actual cost of a similar earlier project	Fast; similarity and expert judgment must be credible
Bottom-up	Estimate detailed activities/work packages, then sum	Detailed and accountable; time-consuming and vulnerable to omissions
Parametric	Apply a validated model using quantities and rates	Scalable; rate, driver, and historical data must fit
Three-point	Consider optimistic, most likely, and pessimistic costs	Makes uncertainty visible; state the averaging/model assumptions

Original worked examples

Parametric: 400 accounts x $90 per migration + $6,000 fixed setup = **$42,000**. Separate fixed and variable costs and avoid counting the same item twice.

PERT-weighted three-point: with O = $30,000, M = $45,000, P = $90,000, the stated weighted mean is (O + 4M + P) / 6 = **$50,000**. A simple average would be $55,000. The phrase three-point alone does not require one universal weighting.

Common IT estimating problems include rushed estimates, weak historical data, inexperienced estimators, optimism, omitted integration/testing, and pressure to present a desired number as an estimate. Use reviews, documented assumptions, ranges, and updates rather than false precision.

CHAPTER 7

Building an estimate: the Surveyor Pro lesson

Textbook pp. 299-304; Figures 7-2 and 7-3; slides 19, 22-25.

Textbook case: Surveyor Pro

The project develops and deploys handheld information tools for government surveyors, building on a prior feasibility/prototype effort. The case covers 100 devices and user training, purchased servers/software, software development, testing, management, and reserves. Its estimates are organized by WBS category, with assumptions explaining the numbers.

Figure 7-2 is a cost model, not just a final dollar amount. Read each row as quantity, rate, resource allocation, and resulting cost; examine the percentage of the total in each major category. **Figure 7-3** compares labor-based and function-point software-development estimates. In the case, $594,000 and $567,000 are close enough under the stated cross-check rule, so the team uses the higher labor estimate. The supporting assumptions make that choice reviewable. Figure 7-2 labels reserves as 20% of the total, but its arithmetic adds **20% of the pre-reserve subtotal**, which is about 16.7% of the final total; always identify the percentage base.

A practical estimating sequence

1. Clarify how the estimate will be used and how precise it must be.
2. Confirm scope/WBS, schedule, resource needs, assumptions, constraints, and risks.
3. Estimate quantities, rates, and effort; distinguish internal resources from contractors.
4. Include management, interfaces, testing, training, deployment, and support where in scope.
5. Explain contingency and the treatment of management reserve; avoid duplicate allowances.
6. Review assumptions and major category proportions with knowledgeable people.
7. Approve the estimate, then time-phase it to create the baseline and funding needs.

Original model fragment

Item	Calculation	Estimated cost
Configuration	160 hours x $75	$12,000
Device purchase	30 devices x $600	$18,000
Training	Fixed supplier proposal	$5,000
Subtotal before reserves	Sum of the three distinct items	$35,000

Basis of estimates should identify the scope version, quantities and rates, estimating methods, exclusions, constraints, risks, and confidence or range. When the device count changes, the model shows what must be updated.

FTE lesson from ResNet/Table 7-2: staffing estimates separate departments and contractors over time. FTE measures equivalent full-time effort, not the number of named people. Software size may also be modelled through function points, user stories, and internal logical files (ILFs), but local evidence must calibrate the conversion to labor and cost.

CHAPTER 7

Budget, baseline, reserves, and funding

Textbook pp. 293, 305-306; Figures 7-4 and 7-5; slides 13, 26-27.

A **cost baseline** is the approved, time-phased budget used to measure performance. A list of estimated total costs is not yet a useful time-phased baseline. The project budget includes the baseline plus management reserve.

Component	Original amount	Treatment
Estimated planned work	$180,000	Starting cost estimates
Contingency reserve	$20,000	Included in the cost baseline
Cost baseline	$200,000	Work plus contingency
Management reserve	$15,000	Outside the baseline
Total project budget	$215,000	Baseline plus management reserve

Figure 7-5 shows the build-up from activity/work-package estimates through control accounts and the baseline to the total budget. Reserve allocations can appear at different aggregation levels; do not add the same reserve repeatedly as it rolls up.

Read Figure 7-4 across time

The Surveyor Pro cost baseline allocates costs by WBS item and month. Staff costs recur; devices, development, testing, and training occur in different periods. Column totals show monthly needs; cumulative totals become the planned-value curve. Rounded displayed entries may not sum exactly to the printed total, as the figure's note explains.

Original funding example

A $100,000 baseline has monthly needs of $20,000, $35,000, and $45,000. If only $40,000 is available by the end of month two, cumulative planned spending of $55,000 creates a $15,000 funding gap. The project may need earlier funding or an authorized schedule/scope adjustment, even though the total approved money eventually reaches $100,000.

Read FTE with a time period

Two people each working 20 project hours in a 40-hour week provide **1 FTE for that week**. At $70 per hour, their 40 total hours cost $2,800. Four weeks at the same effort would be 160 hours. The text's shorthand about 40 hours requires this weekly context; an FTE is not universally 40 hours total.

Trap: management reserve is not a fund the team should quietly spend or move into the baseline. Its use follows the organization's authorization and change-control rules.

CHAPTER 7

Earned value: three views of the same work

Textbook pp. 307-310; Tables 7-3 and 7-4; slides 29-34.

Earned value management (EVM) integrates scope, schedule, and cost. It compares the work scheduled, work actually achieved, and money actually spent at the same status date. Spending below the plan can mean less work was done; spending alone is not enough to judge performance.

Term	Ask this question	Source of the value
PV - planned value	What budgeted work should be done by now?	Approved time-phased baseline
EV - earned value	What is the budgeted value of work actually achieved?	Achieved work using agreed measurement rules
AC - actual cost	What has that work actually cost?	Actual cost records for the same reporting scope/date
BAC - budget at completion	What is the total authorized budget for the measured work?	Total performance baseline budget

Original work-package example

A package has a $40,000 budget. By today, 50% was scheduled and 40% is achieved under the agreed completion rule. Actual spending is $18,000. - PV = 0.50 x $40,000 = **$20,000**. - EV = 0.40 x $40,000 = **$16,000**. - AC = **$18,000**. Actual spending does not determine EV.

If it were actually 60% complete, EV would be $24,000, which exceeds today's PV of $20,000. This is possible because work is ahead of plan. EV still cannot exceed the package's $40,000 total authorized budget merely because spending rises.

Textbook example: server installation

Tables 7-3 and 7-4 use PV $10,000, EV $5,000, AC $15,000 after week one. Only half the planned budget value has been achieved, yet actual costs already exceed the whole planned package cost. CV is -$10,000, SV is -$5,000, CPI is about 0.33, and SPI is 0.50. This shows why comparing AC with PV alone misses the degree of poor performance.

Progress terminology: rate of performance (RP), listed among the chapter key terms, expresses actual work completion/progress; distinguish it from planned completion and money spent.

Measurement caution: hours consumed are not automatically evidence of completed deliverables. Define completion rules, keep dates and scope consistent, and use credible estimates. A 0/50/100 rule is a simplified progress convention, not a law describing exact physical completion.

CHAPTER 7

Earned-value calculations: work and interpret

Textbook pp. 308-310; Table 7-4; slides 30-34. All numbers below are original.

Suppose **PV = $60,000, EV = $48,000, AC = $54,000, BAC = $120,000**. Use amounts cumulative to the same status date.

Measure	Formula and calculation	Interpretation
Cost variance, CV	EV - AC = 48,000 - 54,000 = -$6,000	Work achieved cost $6,000 more than its budgeted value
Schedule variance, SV	EV - PV = 48,000 - 60,000 = -$12,000	Achieved work is $12,000 of budget value below scheduled work
Cost performance index, CPI	EV / AC = 48,000 / 54,000 = 0.8889	About $0.889 of budgeted work earned per $1 spent
Schedule performance index, SPI	EV / PV = 48,000 / 60,000 = 0.80	80% of the scheduled budget value has been earned

Four interpretation rules

- Negative CV is unfavorable cost performance; positive CV is favorable.
- Negative SV is unfavorable schedule progress; positive SV is favorable.
- CPI below 1 is unfavorable; above 1 is favorable cost efficiency.
- SPI below 1 is behind planned work; above 1 is ahead of planned work.

Units matter: CV and SV are expressed in money or budget-value units; CPI and SPI are ratios without units. An SV of -$12,000 does not directly mean 12 days late. Look at the actual schedule and critical path for calendar implications.

Keep cost and schedule separate

A project can be ahead of schedule but over budget: SPI = 1.15, CPI = 0.90. Another can be behind schedule but cost-efficient: SPI = 0.85, CPI = 1.10. Neither index determines the other.

Memory aid: these four formulas all start with EV. Variances subtract; indexes divide. Cost uses AC; schedule uses PV.

Try before checking: EV $75,000, AC $60,000, PV $90,000 gives CV **+$15,000**, SV **-$15,000**, CPI **1.25**, and SPI **0.8333**. The work achieved is cost-efficient but behind the scheduled amount.

CHAPTER 7

Forecasting: EAC, ETC, and required future efficiency

Textbook pp. 308-311; Figure 7-6; slides 31-34.

A forecast is a conditional prediction. Its formula should match the assumption. The chapter's **EAC = BAC / CPI** assumes current cost efficiency continues; it is not a universal forecast valid under every future scenario.

Using the previous example, CPI = 48,000 / 54,000 = 0.888888... and BAC = $120,000: - **EAC = BAC / CPI = $135,000**: forecast total project cost. - **ETC = EAC - AC = $81,000**: forecast additional cost remaining. - The forecast is $15,000 above BAC. Retain full precision until the final result.

TCPI: what must happen next?

To-complete performance index (TCPI) asks what future cost efficiency is required to reach a target, rather than reporting the efficiency achieved so far.

Target	Formula	Original example
Original BAC	(BAC - EV) / (BAC - AC)	(120,000 - 48,000) / (120,000 - 54,000) = 1.0909
Current EAC	(BAC - EV) / (EAC - AC)	72,000 / 81,000 = 0.8889

To finish within the original BAC, the remaining work must earn about $1.091 of budget value per future dollar, compared with current efficiency of about $0.889. That gap needs a credible explanation and action plan. If the denominator is zero or negative, do not blindly interpret the ratio as a normal feasible efficiency requirement.

Time forecast and its limit

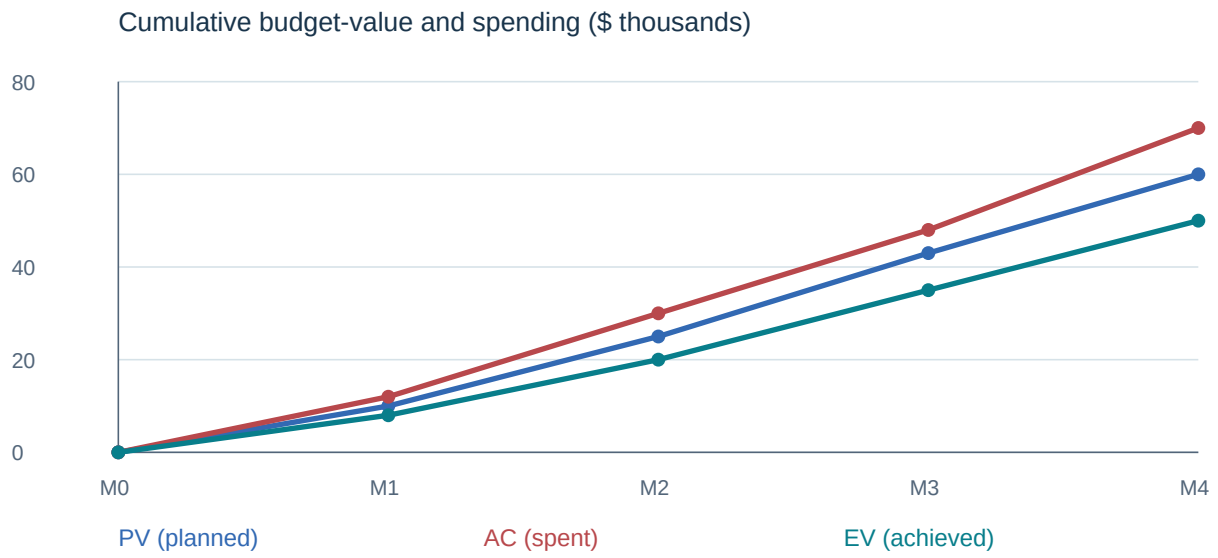
The book also uses **original total duration / SPI**. A 10-month plan at SPI 0.80 gives **12.5 months total**, assuming that efficiency continues. This is not 12.5 additional months. Conventional SPI measures value-based progress, and it tends to 1 when all original scope is earned, even if completion was late. Use schedule analysis alongside the shortcut.

Figure 7-6: PV runs to the original BAC/date; EV and AC stop at the current data date. The EAC point is a forecast. A higher forecast cost and later forecast date signal an expected overrun. Historical actuals and forecasts should remain distinguishable.

CHAPTER 7

Reading an earned-value picture

Textbook pp. 309-311; Figure 7-6. The chart below is an original illustration.



The curves are cumulative values at monthly status dates. At month four, **AC = $70,000, PV = $60,000, and EV = $50,000**. EV lies below both other curves.

Vertical gap between EV and AC: CV = 50,000 - 70,000 = **-$20,000**. The work achieved cost more than its budgeted value.

Vertical gap between EV and PV: SV = 50,000 - 60,000 = **-$10,000**. Less budgeted work has been achieved than was scheduled by this date.

What the chart cannot prove: it does not identify which activity is critical, whether the accepted scope is useful, or whether the remaining plan is realistic. Ask for underlying WBS progress, actual-cost accuracy, approved scope changes, and risk information.

When EVM information is weak

Poor baselines, subjective completion claims, stale actuals, and inconsistent scope/date boundaries can produce misleading outputs. Simplifying the reporting level or completion rules can help, but should preserve a deliverable-oriented WBS and consistent measurement. A sophisticated dashboard cannot repair missing evidence.

Decision example: AC is $35,000 and PV is $40,000, which initially looks inexpensive. EV is only $20,000, so CPI is 20,000 / 35,000 = **0.5714**. Low spending reflects weak progress, not good cost efficiency.

CHAPTER 7

Cost control, portfolios, software, and agile

Textbook pp. 306-316; slides 28, 35-39.

Cost control monitors performance, forecasts outcomes, investigates variance, manages appropriate baseline changes, and informs stakeholders. A forecast can change without authorizing a new baseline. Do not erase a variance by rewriting the original plan to match actual results.

Portfolio decisions go beyond individual cost performance

An efficient project can still be a poor investment if its benefits are obsolete or duplicated. An inefficient project may require corrective action rather than immediate cancellation if its future benefits still justify it. Evaluate future costs/benefits, risk, strategic fit, resource conflicts, and dependencies.

The text's five portfolio-management levels progress from a common project database, to prioritization, investment categories, an automated repository, and risk-return analysis using modern portfolio theory. The historical Schlumberger example illustrates how finding overlapping projects can release funding. It is a portfolio lesson, not a guarantee that purchasing a PPM tool yields a particular return.

What software helps with

Spreadsheets and project/financial systems can assign resource rates, aggregate costs through the WBS, time-phase budgets, report cash flow, calculate EVM, and summarize portfolio health. Project/accounting integration can improve consistency. Staff need training, reliable actuals, and sound coding; automated precision does not establish data validity. References to Project 2016 and old market reports describe the textbook's era.

Adaptive cost management

When scope is uncertain, detailed distant estimates may be wasted. Use high-level forecasts and more detailed near-term estimates. Funding may be secured by release; prioritize the backlog and adjust scope or schedule to stay within a strict cost cap. Agile teams still need financial responsibility.

AgileEVM, as described in the chapter, uses actual costs, estimated backlog size, release iterations, and assumed velocity. The estimates can use hours, points, or another consistent numerical size measure; conversion to earned budget value still needs an explicit model. Story points are not automatically dollars or directly comparable between unrelated teams.

Textbook case conclusion: Juan learns that Surveyor Pro must be justified in financial terms, not only by its technology. Poorly planned projects can lose funding even when their technical ideas are attractive.

CHAPTER 7

Cost chapter: retrieval and application

Based on the chapter objectives, review questions, and exercises, pp. 317-323. Prompts below are original.

Close the guide and answer

1. A sponsor asks for both a cost management plan and a baseline. Explain why one cannot replace the other.
2. Should a $100,000 previous loss determine whether to approve a new $20,000 phase? What information would you use instead?
3. What changes when you move from an estimate to a time-phased budget?
4. How can a project spend less than planned and still have bad cost performance?
5. Why should EAC include an assumption about future performance?

Model answers

1. The plan defines methods, thresholds, and reporting rules. The baseline gives approved budget amounts over time for measurement.

2. The previous loss is sunk and does not by itself justify continuation. Compare the new phase's future costs, benefits, risks, and other genuine future consequences.

3. You allocate estimated costs to WBS work and scheduled periods, approve a performance baseline, and assess when funding is needed.

4. Less work may have been completed. Compare EV with AC, not just AC with PV.

5. Forecasts extrapolate an expected pattern. If the remaining work differs materially, an unqualified BAC/CPI forecast can mislead.

Original calculation drill

At a status date, PV = $100,000, EV = $90,000, AC = $120,000, and BAC = $300,000. Assuming current cost efficiency continues, calculate the four performance measures, EAC, and ETC.

Check: CV = **-$30,000**; SV = **-$10,000**; CPI = **0.75**; SPI = **0.90**; EAC = **$400,000**; ETC = **$280,000**. The project is behind planned work and cost-inefficient. To retain the original BAC, TCPI = (300,000 - 90,000) / (300,000 - 120,000) = **1.1667**.

Revision checklist: explain all four processes; cost classifications; profit/margin and life cycle costs; reserve treatment; estimate ranges and methods; basis of estimates; FTE; PV/EV/AC/BAC; variances/indexes; conditional forecasts; TCPI; software limitations; portfolio and adaptive cost decisions.

CHAPTER 8

Quality: needs, requirements, and three processes

Textbook pp. 328-332, 334-336; slides 4-8, 11, 13.

Quality concerns satisfying the needs for which the project exists. **Conformance to requirements** means meeting written specifications. **Fitness for use** means the result works for its intended purpose. Both matter: a product can satisfy an incomplete specification yet disappoint users.

Original example: an attendance tool includes every specified screen, but instructors cannot correct an entry before marks are finalized. The build may conform to the written screen list while failing a key use. Discuss implied needs early and translate them into clear requirements and acceptance criteria.

Process and group	Main focus	Important outputs
Plan Quality Management - Planning	Define standards, metrics, and how to meet them	Quality management plan; quality metrics; plan/document updates
Manage Quality - Executing	Put the plan into operation and improve design/processes	Quality reports; test/evaluation documents; change requests; updates
Control Quality - Monitoring and Controlling	Measure specific results against criteria	Quality control measurements; verified deliverables; performance information; change requests; updates

Figure 8-1 is the formal process map. Manage Quality includes quality assurance activities plus design and process improvement. The chapter summary and final lecture slide still use some older quality-assurance process wording; use the formal names above when asked to identify a process.

Quality control and acceptance

The narrative groups practical outcomes into **acceptance decisions, rework, and process adjustments**. Rework repairs rejected output; process adjustments reduce recurring problems. These categories explain what the team does after measurement.

Keep the formal handoff distinct: **Control Quality verifies deliverables; Validate Scope formalizes customer acceptance**. The text's p. 336 wording about accepted/validated deliverables is loose compared with Figure 8-1 and Chapter 5. Passing a technical test does not automatically prove the sponsor has formally accepted the scope.

Quality is a shared responsibility. The project manager leads quality management; customers clarify needs, managers supply an enabling system, and team members build and check the work. Late inspection cannot compensate for every upstream failure.

CHAPTER 8

Plan quality as measurable behavior

Textbook pp. 331-334; slides 9-10.

Quality planning anticipates problems and designs methods to prevent them. Relevant inputs include quality policies, scope and product descriptions, stakeholder needs, and applicable standards. The result should say what is measured, under what conditions, by whom, how often, and what counts as acceptable.

Aspect	What it means	Original requirement example
Functionality	Performs the intended functions	Authorized staff can register a student and reverse an incorrect entry
Features	Particular user-valued characteristics	Search filters and contextual help
System outputs	Screens and reports produced	Monthly totals use the agreed date range and reconciliation rules
Performance	How well it handles intended use	95% of searches finish in 2 seconds with 500 simultaneous users
Reliability	Operates as expected under normal conditions	Defined availability and failure behavior during normal service hours
Maintainability	Ease of upkeep, repair, and change	A trained support analyst can complete a documented recovery procedure

Metric means a defined measurement. A threshold without a measurement method can still be ambiguous. For availability, specify the measurement window, treatment of planned maintenance, and what counts as an outage. For response time, specify the workload, dataset, and percentile.

Design of experiments

This technique varies factors systematically to discover which influence the outcome and which combinations work best. For example, test combinations of server memory and configuration under controlled loads. It can reveal interactions that one-factor-at-a-time trial and error misses. The chapter also applies this thinking to staffing and cost/schedule trade-offs.

Textbook case: executive information system (EIS)

A medical-instruments company has a system with access complaints, delays, and supposedly inconsistent reports. Consultant Scott Daniels investigates before recommending replacements. The opening symptoms do not establish their cause. Define the real use, operating conditions, training needs, and quality criteria before assuming that more powerful hardware is the solution.

Trap: adding appealing features does not compensate for unreliable basic functionality. Balance scope, schedule, cost, and the customer's actual quality needs.

CHAPTER 8

Manage quality: improve the way work happens

Textbook pp. 334-335, 356, 359; slides 11-12, 44, 49-50.

Quality assurance gives confidence that the process will satisfy relevant standards. **Manage Quality** also covers design and process improvements. **Control Quality** checks actual results. In a scenario, ask whether the team is defining the approach, improving/assuring the way work is performed, or testing a specific output.

Approach	Main idea	Original application
Quality audit	Structured review of quality activities and compliance; internal or independent	Review whether the team actually follows its peer-review and test procedures
Benchmarking	Compare practices or characteristics with other projects/products	Compare help-desk resolution patterns with another service team
Kaizen	Continuing improvement	Make small evidence-based refinements to defect triage
Lean	Increase customer value while reducing waste	Remove duplicate approvals that create delay without reducing risk
Kanban	Visualize and improve flow while limiting work in progress	Limit simultaneous review items so defects are resolved before more work piles up

The lecture's five Kanban properties are visual workflow, work-in-progress limits, measuring/managing flow, explicit policies, and using models to identify improvement opportunities. Teams need not use identical boards; the method improves their existing context.

Why an audit is not the same as a test

An audit may discover that no one consistently reviews security requirements, even before a defect appears in the release. A test may reveal a specific authorization defect. Both are useful: one examines the way work is managed; the other checks a result.

Leadership and expectations

Top management supplies commitment, resources, learning opportunities, and the conditions for doing the work well. Repeatedly demanding speed while withholding time for reviews can create the defects management later blames on staff. Ask what different customers, departments, and cultures mean by acceptable quality; avoid assuming shared expectations.

Textbook lesson from ResNet: meaningful measurement follows business goals, such as ticket sales and call-handling performance, rather than focusing only on technical activity counts. A metric should support the reason the project exists.

CHAPTER 8

Choose the right quality tool

Textbook pp. 336-342; Figures 8-2 to 8-9; slides 14-22.

Tool	Best question it answers	Main caution
Cause-and-effect / fishbone / Ishikawa	What possible root causes might explain this problem?	A proposed branch is a hypothesis, not proof
Checksheet / tally sheet	How many occurrences happened in defined categories?	Agree on counting and category definitions
Histogram	What does the frequency pattern or distribution look like?	Frequency does not establish the cause
Pareto chart	Which categories contribute the most counted problems?	Frequency is not the only priority; severity matters too
Scatter diagram	Do two variables appear related?	Association does not prove causation
Flowchart	What steps and decisions make up the process?	Map the real process rather than an idealized policy
Run chart	How does a measure change over time?	A time pattern alone does not supply statistical control limits
Control chart	Is process behavior stable, or does it show special-cause signals?	Stability and meeting customer specifications are different

The assigned list is called the **Seven Basic Tools**, with flowcharts/run charts discussed together and stratification mentioned. Recognize the purpose of each rather than getting trapped by variations in how references count the tools.

Reading the textbook figures

Figure 8-2 groups EIS access-problem causes into hardware, software, and training. Figures 8-4 to 8-6 show complaint collection, relationships, and frequencies. Figure 8-7 ranks complaint types; Figure 8-8 traces a deliverable decision process; Figure 8-9 shows defect trends over time.

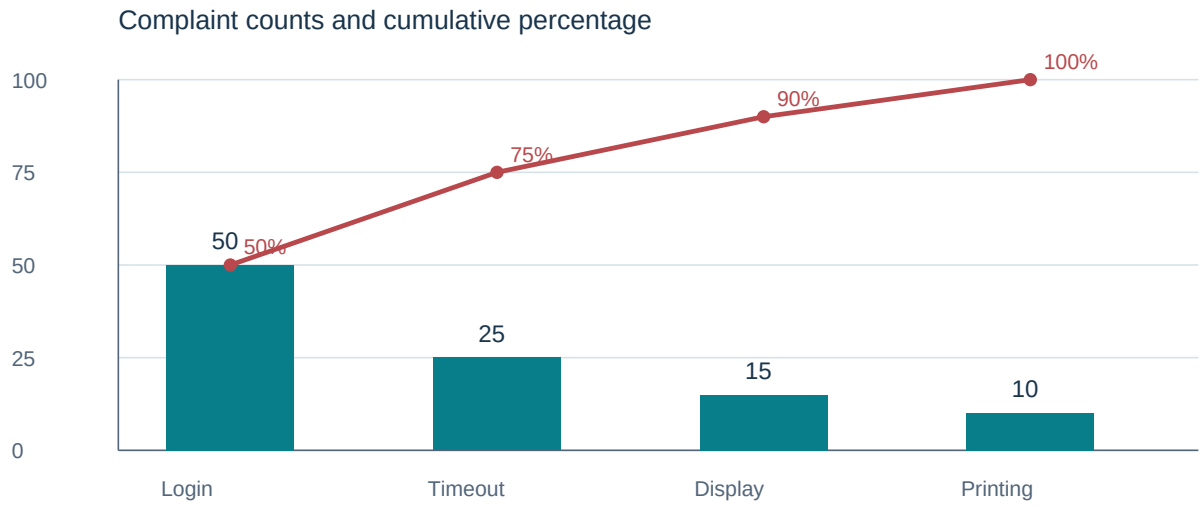
Five whys means repeatedly asking why a symptom occurs to work toward a cause. Five is a useful prompt, not a mandatory count or a guarantee that the first explanation is true.

Scenario: a team suspects response time rises with load. Use a scatter plot to inspect association; then use controlled tests and technical investigation. If the question instead asks which complaint category to tackle first, a Pareto chart is more directly relevant.

CHAPTER 8

Pareto analysis: rank, add, investigate

Textbook pp. 340-341; Figure 8-7. The data below are original.



Bars: counts (left scale). Line: cumulative % (same 0-100 numerical scale).

A help desk records **50 login**, **25 timeout**, **15 display**, and **10 printing** complaints. There are 100 complaints, so the first two categories account for **75%**. Bars are ranked from largest to smallest; the cumulative line helps show the share addressed by tackling several categories.

Interpret the 80/20 idea correctly

The Pareto principle says a relatively small group of causes often accounts for a large share of problems. It does not require every dataset to be exactly 80% and 20%. Calculate the actual percentages.

Frequency is not the only decision criterion

A rare defect exposing confidential records may deserve urgent action even if login complaints are more numerous. Consider severity, regulatory implications, cost, and business impact alongside the counts. A Pareto chart prioritizes investigation; it does not prove that the highest bar identifies a single root cause.

Use the tools together

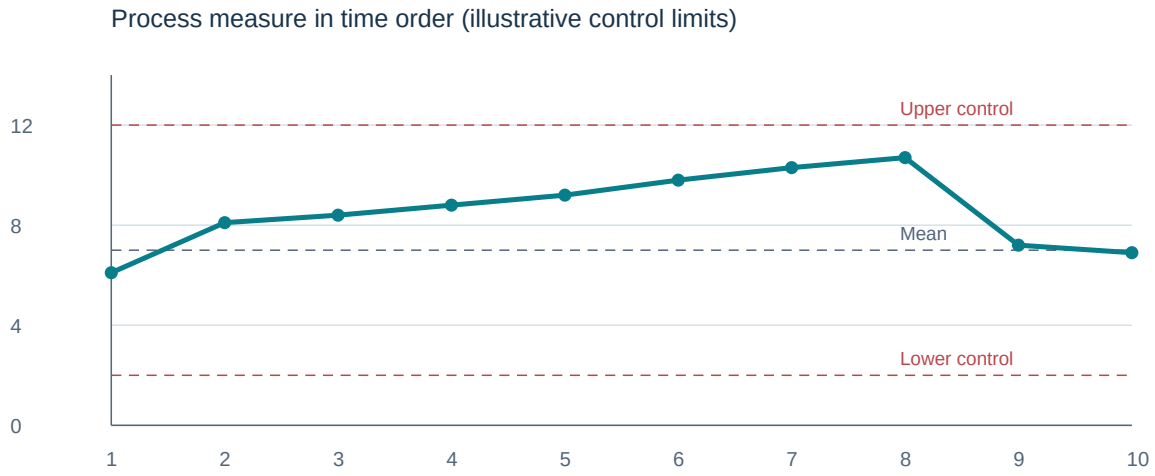
Start with a checksheet to collect consistently defined categories. Create a Pareto chart to locate major contributors. Use a fishbone and evidence to investigate causes. Pilot a change, then use time-based measures to check whether performance improves.

Textbook EIS connection: login problems and lockups dominate the complaints in Figure 8-7. The later case investigation finds training problems rather than major hardware/software faults. The complaint label alone did not reveal the solution.

CHAPTER 8

Control charts, run charts, and specification limits

Textbook pp. 337-338, 342; Figures 8-3 and 8-9; slide 16. Chart below is original.



Observations 2-8 rise consecutively: investigate the seven-point pattern.

Control limits describe expected process behavior from statistical analysis. **Specification limits** describe the outcomes the customer requires. They answer different questions. A stable process can consistently fail the customer's requirements; a single acceptable output does not demonstrate process stability.

The assigned seven-run rule

The textbook flags **seven successive observations** all above the mean, all below it, or consistently increasing/decreasing. This is a signal to investigate nonrandom behavior even when individual observations remain inside the limits. In the illustration, the seven middle observations rise; investigate the pattern rather than dismissing it because no value crossed a line.

Do not overreact to every fluctuation. Random/common-cause variation is inherent in a stable process. Changing settings after every small change can make behavior worse. A special-cause signal needs investigation; the appropriate response depends on the evidence.

Read Figure 8-3 carefully

The ruler example shows different control and customer-specification limits. Some points can violate the pattern rule without violating specifications. The figure tests your ability to distinguish trend, stability, and conformance.

Run chart versus control chart

A run chart plots measurements in time order and is useful for spotting trends. A control chart adds the statistically derived reference structure needed to assess process stability. Both preserve time order; a Pareto chart instead sorts categories by frequency.

Applied answer: seven rising response-time observations suggest a possible process shift. Investigate causes such as changing load or configuration. The pattern alone does not prove seven defective transactions, and statistical stability alone would not prove response times are acceptable.

CHAPTER 8

Statistical sampling: calculate without confusing the inputs

Textbook pp. 342-343; Table 8-1; slides 23-24. Statistical clarification: NIST, Selecting Sample Sizes.

Statistical sampling inspects part of a population when inspecting every item is impractical. A good design must make the sample relevant to the population; a large but biased convenience sample can still mislead. Define what is being estimated and how items are selected.

The chapter's simplified proportion-based expression is **$n = 0.25 \times (z / e)^2$**. Here z is the certainty factor, e is the desired acceptable error or margin of error expressed as a decimal, and n is the calculated sample size. The 0.25 term corresponds to the conservative p(1-p) value at p = 0.5; the approximation has assumptions and is not a universal sample-size formula for every measurement.

Confidence used in the chapter	z value	Example margin of error e
95%	1.960	0.05 if the desired precision is 5 percentage points
90%	1.645	0.10 in the book's example
80%	1.281	0.20 in the book's example

Original worked calculation

For 95% confidence and e = 0.05: $n = 0.25 \times (1.96 / 0.05)^2 =$ **384.16**. If asked for the minimum whole-number sample meeting this calculation, **round up to 385**. The textbook displays approximately 384. For z = 1.645 and e = 0.10, the calculation is 67.650625, so use **68** when rounding up.

Important textbook simplification

The prose on p. 343 ties acceptable error to 1 minus the confidence level. Statistically, **confidence and margin of error are separate choices**; a 95% confidence target does not force e to equal 0.05. Use the explicit error specified in a question. This clarification follows the NIST sample-size discussion, which treats precision and confidence/risk separately.

Holding other inputs fixed, halving e makes n four times larger. Raising confidence usually raises the required sample. Sampling reduces inspection effort; it does not guarantee that every rare defect or unusual requirement will be found.

CHAPTER 8

Six Sigma, DMAIC, and variation

Textbook pp. 344-348; Tables 8-2 and 8-3; slides 25-33.

Six Sigma is a customer-focused, data-based approach to improving processes and business results. Its conventional target is about **3.4 defects per million opportunities (DPMO)**. It combines leadership commitment, trained teams, disciplined methods, and project selection; it is more than a slogan for near-perfect inspection.

DMAIC phase	What the team does	Original example
Define	Establish the gap, customer needs, and problem scope	Invoices reach customers with incorrect tax codes
Measure	Define measures and collect baseline data	Count defects and their opportunities consistently
Analyze	Investigate and verify root causes	Check whether a rule/configuration mismatch predicts errors
Improve	Develop, pilot, and refine a solution	Trial a validated tax-code check
Control	Sustain and monitor the improvement	Track the revised process and respond to new signals

A suitable improvement project has a meaningful performance gap with causes and best solution not already fully known. The book's phrase about an unclear problem should not mean a vague charter: define the observed gap clearly while investigating its underlying cause. A predetermined software purchase with no diagnosis is a weak DMAIC starting point.

Defects are not the same as defective units

One form can have several defect opportunities. For 2,000 forms, 5 opportunities each, and 20 defects, DPMO = 20 / (2,000 x 5) x 1,000,000 = **2,000**. Define opportunities consistently; changing the denominator changes the interpretation.

Read the two sigma tables as different conventions

Table 8-2 describes a centered normal distribution: about 68.3%, 95.5%, and 99.7% of values lie within 1, 2, and 3 standard deviations. Its centered six-sigma tail is not the same as Table 8-3's conventional Six Sigma program conversion. The latter allows for long-term process variation; do not equate the tables' tail probabilities.

Six nines means 99.9999%, distinct from the Six Sigma convention. For availability over a 365-day year, the unavailable fraction gives about **31.54 seconds**, rounded to roughly 30 seconds in the book. Belt labels describe training/responsibility, from Yellow and Green through Black and Master Black Belt; exact training durations are contextual textbook descriptions.

CHAPTER 8

Testing throughout development

Textbook pp. 348-350; Figure 8-11; slides 34-37.

Testing should be planned and performed across the life cycle. Waiting until the end makes defects harder to diagnose and more expensive to repair. Reviews, good requirements, design checks, and disciplined development complement testing rather than being replaced by it.

Test level	Main target	Original scenario
Unit	A single component in isolation	Check that a tax calculation function handles a boundary value
Integration	Interfaces between grouped components	Check that the checkout module passes the right format to payment processing
System	The full system as an integrated whole	Exercise an end-to-end transaction under the specified environment
User acceptance (UAT)	Fitness with users' business needs before acceptance	Staff confirm they can complete real return/refund tasks

Figure 8-11 connects requirements, architecture, build/buy decisions, implementation, and testing tasks. Unit testing precedes integration and system testing, with UAT assessing business fit. The important lesson is to plan tests alongside the relevant development work; the figure is not a reason to postpone all test thinking until coding ends.

Other test purposes

Performance tests examine behavior such as response time and throughput under specified conditions. Scalability tests examine how behavior changes as demand grows. Alpha and beta testing are additional testing arrangements mentioned in the chapter. A test's level and its purpose are different dimensions; a full-system test can also be a performance test.

Why passing tests is not proof of no defects

Complex systems have enormous combinations of inputs, states, integrations, and user behavior. A finite test suite checks selected conditions. Humphrey's argument is that developers should prevent, measure, and learn from their errors, with management support, rather than rely only on repeated code/test/fix cycles.

Application: modules passing independently but failing when connected calls for integration testing. Technical tests passing while users cannot complete their work points toward a business-fit/requirements gap that UAT should expose.

Cost link: finding a pre-delivery defect creates internal failure/rework cost; finding it only after customer release creates external failure exposure. Retesting a repaired output is appraisal work, distinct from the repair itself.

CHAPTER 8

Quality thinkers: the associations and the reasoning

Textbook pp. 351-355; slides 38-41.

Thinker	Contribution emphasized in this edition	What it means in practice
Deming	Management system, 14 points, continuing improvement; plan-do-check-act	Improve the process and conditions instead of relying on final inspection and blame
Juran	Fitness for use; management commitment; planning, control, and improvement trilogy	Define customer needs and organize measurable improvement
Crosby	Zero defects; Quality Is Free	Doing work correctly avoids costly nonconformance; prevention can pay for itself
Ishikawa	Fishbone diagrams and quality circles	Involve people doing the work in studying and improving it
Taguchi	Robust design and engineered experiments	Design quality in and reduce deviation from the target
Feigenbaum	Total quality control; quality responsibility with the people doing the work	Support organization-wide quality rather than assigning it only to inspectors

Paraphrased map of Deming's 14 points

1. Maintain a lasting purpose of better products and services.
2. Adopt a management philosophy that values quality.
3. Build quality in rather than depend on final inspection.
4. Choose supplier relationships for total value, not the lowest price alone.
5. Keep improving the system of work.
6. Train people for their jobs.
7. Lead and support people effectively.
8. Make it safe to raise problems.
9. Remove barriers between departments.
10. Replace empty slogans with real methods and resources.
11. Avoid quotas and numeric demands unsupported by capable processes.
12. Remove conditions that undermine pride in good work.
13. Support education and self-development.
14. Involve the organization in sustained improvement.

Interpretation: these principles favor management responsibility, prevention, learning, and cooperation. They do not mean measures are useless; measures need a sound purpose and process context.

Quality circles are voluntary groups studying improvements in their own work area. **Robust design** aims for reliable performance despite variation, rather than depending on sorting failures out at the end.

CHAPTER 8

Improvement programs, standards, and leadership

Textbook pp. 352-359; slides 39-50.

Juran's ten-step improvement sequence, paraphrased

Build awareness; set goals; organize improvement teams and responsibilities; train; carry out improvement projects; report progress; recognize contributions; communicate results; track performance; and embed improvement in regular organizational routines. His trilogy connects planning the desired quality, controlling results, and improving performance.

Crosby's improvement sequence, paraphrased

Secure management commitment; form cross-functional teams; measure quality problems; evaluate quality costs; raise awareness; act on problems; organize a zero-defect effort; train supervisors; mark the commitment publicly; set improvement goals; remove barriers; recognize contributions; keep quality councils active; and repeat the cycle. Zero defects is a performance aspiration, not evidence that testing has established mathematical perfection.

Baldrige and ISO

The textbook presents the **Malcolm Baldrige National Quality Award** as recognition of organizational excellence and quality/performance management, rather than approval of a particular product. Its historical seven assessment areas include leadership, strategy, customer focus, information/analysis, people, processes, and results. Award categories, counts, and wording in the slides describe that historical presentation, not a current application guide.

The **ISO 9000 family** supports systematic quality management through consistent planning, control, documentation, and improvement. Use the assigned textbook's framing for this course. It should not be read as proof that a certified system guarantees every delivered product has no defects. Names, editions, and certification requirements should be checked separately if making a real compliance decision.

Leadership and the workplace

The DeMarco/Lister study in the chapter reports much greater variation between organizations than between people in the same organization. Its lesson is that work conditions and organizational practices matter: give people suitable space, clear direction, fewer political obstacles, and the ability to do careful work. Do not turn one historical study's correlations into universal guarantees about languages, pay, or experience.

Management action: when rushed, ambiguous requirements create repeated rework, improve the upstream definition and review process and protect focused work time. A new testing tool alone is unlikely to solve the whole problem.

CHAPTER 8

Cost of quality: spending now versus failing later

Textbook pp. 356-357; slides 45-46.

Cost of quality = cost of conformance + cost of nonconformance. Conformance includes preventing and assessing defects; nonconformance includes failures and their consequences. The objective is the best total outcome for the required quality, not minimizing every category separately.

Category	Meaning	Original example
Prevention	Plan and perform work to avoid defects	Training, better requirements/design methods, supplier qualification
Appraisal	Evaluate processes or output for quality	Inspection, test execution, test-result analysis
Internal failure	Correct defects before the customer receives the output	Rework of a faulty release before delivery
External failure	Address defects that escaped to the customer	Warranty repair, complaints, liability, lost future business
Measurement/test equipment	The text separately identifies capital equipment supporting prevention/appraisal	A dedicated test rig

Boundary that matters: internal versus external failure depends on whether the customer has received the defective output. It does not mean whether the person repairing it works inside the company. A contractor can fix an internal failure before delivery; an employee can repair an external failure after delivery.

Original investment example

Current release costs include $12,000 appraisal, $25,000 internal failure, and $40,000 external failure: **$77,000 total**. A proposed $8,000 prevention program keeps appraisal at $12,000 and is expected to reduce internal and external failure to $10,000 and $15,000. New total = 8,000 + 12,000 + 10,000 + 15,000 = **$45,000**, an expected **$32,000 reduction** under the stated assumptions.

Classification trap: repair versus retest

Fixing a pre-delivery defect is an internal failure cost. Testing the corrected product is an appraisal activity. If a case bundles the two, ask what expenditure the question is classifying; do not assume every task that occurs after finding a defect is identical.

Equipment trap: if capital test-equipment costs are reported separately, do not also include the same amount in appraisal and add it again. The textbook lists five categories; other summaries often group the first four and allocate equipment to supporting activities.

CHAPTER 8

Maturity models and disciplined improvement

Textbook pp. 359-362; slides 51-54. Level labels follow the supplied edition.

A **maturity model** is a framework for assessing and systematically improving processes. It is about repeatable organizational capability, not just whether one project had a successful outcome.

SQFD - Software Quality Function Deployment translates user requirements into prioritized, measurable technical specifications. Introducing quality during requirements and design can reduce expensive downstream changes.

Assigned CMMI level descriptions

Level	Assigned label	Main distinction
0	Incomplete	Work is missing or only partly performed
1	Performed	The process does the required work, but may not be institutionalized
2	Managed	Planned and controlled work has people, resources, and policy support
3	Defined	Projects tailor documented organizational standard processes
4	Quantitatively managed	Quantitative objectives and statistical techniques control performance
5	Optimizing	Evidence about variation drives continuing and innovative improvement

The book describes this 0-5 list as capability levels, while the discussion also uses maturity language. Treat it as the **assigned list for this edition**. CMMI terminology depends on the representation and version; do not assume that every other source's capability/maturity list has these same labels. This guide does not substitute a modern framework for the assigned content.

OPM3 - Organizational Project Management Maturity Model

The chapter uses OPM3 to connect organizational strategy with better **project, program, and portfolio** practices. Improvement progresses through **standardize, measure, control, and improve**. The historical publication dates and counts of practices/capabilities describe the edition's source material, rather than something to infer about present products.

Example of the hierarchy: a best practice might be supporting an internal PM community; a capability is the ability to facilitate its activities; an outcome is an established local initiative; an indicator provides evidence that the community is addressing real issues. A stated capability without observable results is weak evidence of improvement.

Application: a Level 3-style process uses common defined practices tailored to the project. A Level 4-style process goes further by quantitatively managing performance. Level 5 adds sustained improvement; documentation alone does not establish that level.

CHAPTER 8

Quality in agile delivery and the EIS case conclusion

Textbook pp. 362-364; slides 55-57.

Adaptive quality management builds frequent quality work into delivery rather than postponing it. Small batches expose inconsistencies earlier, when the cost of change is usually lower. Quality is still required; shorter iterations do not create permission to accept unchecked output.

Four complementary practices

- Agree on a **definition of done** that includes the quality criteria required for completed work.
- Test small increments and obtain relevant user feedback on business fit.
- Use **retrospectives** to identify process causes, try improvements, and evaluate whether the changes worked.
- Track **escaped defects**, meaning defects that passed the intended earlier detection point and reached a later stage or the customer. Define the boundary used by the metric.

The book expresses a zero-escaped-defect goal for agile software quality. It is an improvement target, not a guarantee obtained merely by using Scrum. Hybrid projects may need both iteration-level checks and broader system, integration, governance, or release acceptance checks.

Software supports the method

Spreadsheets/charting tools can create quality charts; statistical tools support analysis; test/defect systems track evidence; project tools schedule quality work; specialist tools support Six Sigma and maturity assessment. Choose tools that fit the quality question and keep their data trustworthy.

Textbook case conclusion: Scott Daniels and the EIS

Scott's team uses cause-and-effect analysis, complaint information, and investigation. They find that many users lack appropriate training. They find no major hardware problem, and the alleged inconsistent-report issue is traced to a manager misreading reports rather than a software-design defect.

Why this is an application case: symptoms that sound technical can have a training or expectation cause. Replacing the server before investigating could waste money. Improve the actual cause, document evidence, and incorporate lessons into future project planning.

This outcome does not imply every access or reporting problem is a training issue. The transferable method is to investigate, measure, and verify the explanation before choosing a solution.

CHAPTER 8

Quality chapter: retrieval and application

Based on objectives and chapter reviews, pp. 365-370. Prompts below are original.

Close the guide and answer

1. How can a product meet specifications but still lack quality for the customer?
2. Distinguish a quality audit, an integration test, and formal scope acceptance.
3. Why can seven observations within specifications still call for investigation?
4. What makes customer-side warranty repair an external failure cost?
5. How does a retrospective differ from a review of the latest product increment?

Model answers

1. The specifications may not capture the user's intended work or implied needs. Conformance needs to be combined with fitness for use.

2. An audit examines quality activities/processes. Integration testing checks component interactions; formal scope acceptance is the customer's acceptance decision through Validate Scope.

3. The assigned seven-run rule can flag a nonrandom pattern without individual values crossing specification limits. Stability and conformance answer different questions.

4. The defect has escaped to the customer. The boundary is delivery, not the employer of the technician doing the repair.

5. A retrospective examines how the team works and how to improve it. A product review/feedback activity examines what was delivered and what stakeholders need next; both can inform quality.

Original calculations

Sampling: $z = 1.96$, $e = 0.04$. $n = 0.25(1.96 / 0.04)^2$ = **600.25**, so the minimum whole-number sample is **601**.

DPMO: 12 defects across 1,500 records with 4 opportunities each gives 12 / 6,000 x 1,000,000 = **2,000 DPMO**. This is not 12 defective records unless each defect occurred on a different record.

Revision checklist: quality definitions and process groups; verified versus accepted output; metrics and design of experiments; QA/audits; all tools and their best uses; control/specification limits; sampling; DMAIC and opportunities; Six Sigma versus six nines; test levels; experts and standards; cost of quality; leadership/workplace; maturity; adaptive quality and the EIS case.

BOTH CHAPTERS

Formula sheet and common mix-ups

Textbook pp. 290, 295-298, 305-310, 343, 347-348, 356-357.

Formula or distinction	Use and limit
Profit = revenue - expenditures	Dollar result; margin divides by revenue
Cost baseline + management reserve = project budget	Contingency is already inside the baseline
PERT-weighted cost = (O + 4M + P) / 6	Use only when this weighting is specified
EV = budgeted value of achieved work	Not actual spending; use the agreed measurement rule
CV = EV - AC; SV = EV - PV	Money/budget units; favorable above zero
CPI = EV / AC; SPI = EV / PV	Ratios; favorable above one
EAC = BAC / CPI; ETC = EAC - AC	Assumes current cost efficiency continues
TCPI(BAC) = (BAC - EV) / (BAC - AC)	Future efficiency required for original BAC
TCPI(EAC) = (BAC - EV) / (EAC - AC)	Future efficiency required for current EAC
Total duration estimate = original duration / SPI	Simplified, conditional textbook projection
$n = 0.25(z/e)^2$	Confidence and error separately specified; round up for a minimum
DPMO = defects / total opportunities x 1,000,000	Total opportunities = units x opportunities per unit
Cost of quality = conformance + nonconformance	Include categories consistently without double counting

Six pairs to explain without looking

Estimate / baseline: likely cost versus approved cost distributed over time. **BAC / EAC:** original authorized performance budget versus forecast total. **EAC / ETC:** total forecast versus forecast remaining. **CPI / TCPI:** efficiency achieved versus efficiency required in the future. **Audit / inspection:** review of quality processes versus checking specific output. **Control / specification limits:** statistical process behavior versus customer requirements.

Recommended study priorities, not confirmed exam predictions: earned-value interpretation and calculations; reserve and document distinctions; choosing a quality tool; cost-of-quality classifications; sampling inputs/rounding; test levels; and the difference between conformance and fitness for use. Also review the wider chapter topics so recognition questions do not surprise you.

FINISH HERE

Source notes, discrepancies, and coverage

Scope: the supplied Chapter 7 and 8 lectures and Schwalbe's assigned ninth edition.

Primary course sources

Chapter 7 lecture: 40 PDF slides. **Chapter 8 lecture:** 57 PDF slides. **Schwalbe, Information Technology Project Management, ninth edition:** Chapter 7 content/review pp. 285-323; Chapter 8 content/review pp. 327-370. The supplied textbook viewer is 25 pages ahead of printed numbering.

Coverage was checked against chapter headings, learning objectives, summaries, key terms, review questions, and the lecture decks. The numerical spreadsheet figures and quality charts were read for their purpose and interpretation rather than reproduced. The guide includes Surveyor Pro, ResNet lessons discussed in these chapters, and the EIS case. Practice scenarios and questions are original and are not the textbook quiz or an actual exam.

Source discrepancies worth recognizing

- **Definitive estimate:** slide 17 repeats a budget-planning purpose; textbook pp. 295-296 emphasize detailed purchasing/final-cost estimates. Use the textbook distinction.
- **Quality process name:** the formal process is Manage Quality. Older assurance wording persists in the chapter summary and slide 57; assurance is part of the broader process.
- **Verified/validated wording:** Figure 8-1 lists verified deliverables. Page 336 uses loose acceptance/validated language; keep Control Quality and formal Validate Scope acceptance distinct.
- **Sampling:** p. 343's printed 384 is approximate; 384.16 rounds up to 385 for a minimum whole-number sample. Confidence and acceptable error are separate choices, despite the prose linking them.
- **Seven tools:** the chapter groups some flowchart/run-chart/stratification discussion. Learn the displayed tools' purposes rather than assuming all sources count identically.
- **CMMI:** the 0-5 list and its terminology are preserved as assigned; do not silently merge capability and maturity lists from different representations/versions.
- **EV and PV:** EV can exceed scheduled-to-date PV; it is bounded by the component's total authorized budget, not necessarily today's planned amount.

Historical material and outside clarification

Old project statistics, media cases, award descriptions, software versions, and maturity-model publication facts reflect the edition's period. Their lessons matter more than treating them as current news. The text itself notes weak sourcing for one USB/e-cigarette anecdote; do not rely on it as an established case fact.

For the sampling clarification only, consult NIST's **Selecting Sample Sizes**: https://itl.nist.gov/div898/handbook/ppc/section3/ppc333.htm. NIST distinguishes precision, confidence/risk, population variability, and practical sampling constraints. The rest of this guide follows the supplied course sources.

Next step: use the Chapters 7-8 website practice, explain each miss, then reread the cited pages. No particular grade or assessment coverage is guaranteed by completing one bank.